RAJ MAURYA

Software Developer | Analytical Thinker

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EXPERIENCE

Software Development Intern, *Pinch* May'26–Jul'26
- Shipped and tested production features within an agile engineering team
- Translated product requirements into scoped, reviewable delivery increments
- Validated releases with design and QA, documenting edge cases before handoff

Teaching Assistant, *Himshikhar–IIT Mandi* May'26–Jul'26
- Mentored undergraduate cohorts, translating complex concepts for mixed audiences
- Supported course delivery and assessment under fixed academic deadlines
- Managed competing priorities across concurrent student groups

Research Intern, *IIT Mandi* Dec'24–Jan'25
- Studied NP-complete classes and formal reductions of intractable problems
- Decomposed hard problems into tractable, approximable sub-problems

EDUCATION

B.Tech, Mathematics and Computing 2023–27
Indian Institute of Technology, Mandi

- **CGPA:** 8.73/10
- **Coursework:** Optimization, Mathematical Modeling, ML, Deep Learning, DSA, Linear Algebra, PDEs/ODEs, Probability & Statistics, Computer Networks, Quant Finance

PROJECTS

GenWeb — AI Website Generation Platform Git Web
Next.js · React · Supabase · Clerk · LLM Agents
- Agent-based system turning a natural-language brief into a deployed website
- Automated scaffolding, build and deploy into one prompt-driven workflow
- Collapses a multi-day delivery cycle into a single self-serve request

NetSentinel — Real-Time Threat Detection Platform Git
FastAPI · React · Scapy · Python · Data Visualization
- Live traffic monitoring platform surfacing anomalies and threat signatures
- Operator dashboards turning packet telemetry into prioritised alerts
- Streaming ingestion pipeline feeding a real-time visual analytics layer

MovieFlix — Booking & Operations Platform Git
React · Node.js · Express · MongoDB · Stripe
- Full-stack ticketing system spanning booking journey and admin console
- Seat selection, Stripe payments and real-time confirmation workflows
- Single platform serving two distinct user roles with separate dashboards

C-TAEA — Constrained Multi-Objective Optimization Git
Python · Evolutionary Algorithms · IEEE TEVC Replication
- Implemented a two-archive evolutionary algorithm from the IEEE TEVC paper
- Benchmarked convergence and solution diversity on constrained MOPs
- Evaluates competing objectives under hard feasibility constraints

ONGOING WORK

Nano-fluids Flow via PINNs:- Under Review, Journal of Computational Physics

PINNs for Electo-osmotic Flow:- Under Review, Physics of Fluids, Special Edition

Neuro Trade via PINNs:- Indian market modeling using differential equations

EXOCORTEX:- Knowledge Graph for own Digital Twin

POSITIONS OF RESPONSIBILITY

Member, Research Society Since 2024
IIT Mandi

ABOUT ME

Mathematics & Computing undergraduate who pairs quantitative problem diagnosis with hands-on delivery. Builds AI, data and full-stack systems end to end — from framing the problem to shipping the solution. Evaluates technology on merit rather than preference, and turns technical options into decisions non-technical stakeholders can act on.

ANALYTICAL SKILLS

Problem Diagnosis · Solution Design · Trade-off Analysis · Agile Delivery

TECHNICAL SKILLS

AI & Data
PyTorch · TensorFlow · Keras · Scikit-learn · OpenCV

Engineering & Cloud
Python · C++ · JS · Node.js · React · Next.js · FastAPI · Flask · MongoDB · Git · Linux · Bash

Modeling & Optimization
PINNs · ODEs · PDEs · SDEs · Optimization · Stochastic

Scientific & Quant
MATLAB · COMSOL · Qiskit · Quant Finance · NumPy

CERTIFICATIONS

Machine Learning
Coursera, Andrew NG

Quantum Computing
Coursera

Dynamic Programming
Coursera

KEY ACHIEVEMENTS

- **AMEX CodeStreet 2026 Finalist**
  Top 6 team; Cross-channel Customer Journey stitching platform
- **C-TAEA Research Implementation**
  Full IEEE TEVC paper replication, GECCO-track candidate
- **DeepFake Video Classifier**
  CNN classifier for real vs. manipulated video
- **Portable Oxygen Concentrator**
  IoT-enabled prototype for low-resource settings

LANGUAGES

English · Hindi